Learning Spatial–Temporal Consistency for Satellite Image Sequence Prediction

Kuai Dai, Xutao Li, Chi Ma, Shenyuan Lu, Yunming Ye, Di Xian, Lin Tian, and Danyu Qin

***Abstract*— As an extremely challenging spatial–temporal sequence prediction task, satellite image sequence prediction has various and significant applications in real-world scenarios. Although a lot of deep learning prediction models are developed for spatial–temporal sequence prediction, the methods still deliver unsatisfactory performance in terms of keeping spatial–temporal consistency, which leads to inaccurate and blurry satellite image sequence predictions. To maintain spatial–temporal consistency and achieve high-quality satellite image sequence prediction, we propose a novel and effective spatial–temporal consistency network (STCNet). In STCNet, a multilevel motion memory-based predictor is proposed to accurately predict motion patterns of satellite image sequences to ensure temporal consistency. Then, a time-variant frame discriminator is carefully designed and proposed, which can enhance the perception quality of predicted frames to guarantee spatial consistency and simultaneously maintain the motion coherency of predicted sequences. Moreover, a scheduled sampling strategy is proposed to reduce the optimizing difficulty and better train the proposed method. Comprehensive experiments conducted on satellite image sequences from the FY-4A meteorological satellite verify the effectiveness, applicability, and adaptability of our proposed method compared to state-of-the-art approaches under challenging scenarios.**

***Index Terms*— Deep learning, satellite image sequence prediction, spatial–temporal consistency, spatial–temporal sequence prediction.**

I. Introduction

THE satellite image sequence prediction is a challenging spatial–temporal sequence prediction task, which has many vital applications in real-world scenarios such as weather nowcast [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], lightning prediction [11], [12], landslide disaster prediction [13], and floods prediction [14], [15]. Extrapolating reliable satellite image sequences is beneficial to managing social activities and reducing unnecessary economic losses, which makes the task become a hot research topic in recent years.

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In the past few years, we have witnessed many approaches developed for spatial–temporal sequence prediction. According to the main working principle, the prediction methods are roughly cataloged as two taxonomies, namely, opticalflow methods and deep learning prediction methods [16], [17]. Opticalflow methods [18], [19], [20], [21] make predictions by wrapping the current frame with the motion field estimated from the previous two consecutive frames. This type of methods enables to generate sharp future frames in terms of appearance details but fails to model complicated motion patterns such as accumulation and dissipation, which leads to inaccurate motion predictions. Thus, opticalflow methods cannot serve as mainstream baseline methods for spatial–temporal sequence prediction. As data-driven prediction methods, the deep learning models [22], [23], [24], [25], [26], [27], [28], [29] learn spatial–temporal dynamics by fitting huge historical data under the supervision of the objective loss function such as mean absolute error (MAE) or mean square error (mse). Once trained, the deep learning prediction models produce future image frames according to given historical observations. With advanced neural network architectures, deep learning prediction models can effectively exploit historical observation data and hence much better deal with complicated temporal variations than opticalflow approaches. However, as time step goes by, the uncertainty of future states dramatically increases. Correspondingly, the deep learning prediction models cannot accurately infer the future states and tend to produce average results with the guidance of the loss function (MAE or mse), which leads to blurry appearance details. Overall, compared with the opticalflow approaches, this type of methods can make much more accurate predictions in terms of motion dynamics, while the appearance details tend to be blurry. Worst still, the blurry issue becomes more obvious as the prediction time step increases, which seriously impedes long-term predictions.

To alleviate the blurry issue, some researchers try to apply generative adversarial networks (GANs) [35], [36], [37], [38], [39] into spatial–temporal sequence prediction [2], [5], [7], [30], [31], [32], [33], [34]. Specifically, the GAN-based prediction models contain a predictor and a discriminator. The predictor aims to make predictions based on the input frames, while the discriminator accounts for judging the predicted frame and ground truth. By conducting the adversarial training strategy between the predictor and the discriminator, the perception quality of predicted frames can be enhanced. However, discriminators in the GAN-based methods make few efforts

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TABLE I
CHARACTERISTICS OF EXISTING PREDICTION METHODS FOR SATELLITE IMAGE SEQUENCE PREDICTION IN TERMS OF PRESERVING SPATIAL–TEMPORAL CONSISTENCY

Model	Temporal Consistency	Spatial Consistency
Opticalflow Methods [18]–[21]	✘	✔
Deep Learning Methods without GAN [22]–[29]	✔	✘
Deep Learning Methods with GAN [2], [5], [7], [30]–[34]	✘	✔
STCNet (Ours)	✔	✔

to characterize the temporal change in predicted sequences and equally treat the predicted frames at different time steps. As a result, the motion coherency of predicted sequences always tends to be destroyed. In addition, due to the insufficient spatial–temporal modeling ability of the aforementioned prediction methods, the predictors cannot extrapolate reliable motion patterns. As a consequence, the discriminators tend to amplify the error of predictions rather than enhance the visual details of predicted frames.

To make accurate predictions, a forecast model needs to achieve spatial–temporal consistency between predicted sequences and ground-truth ones, which means the prediction method should make efforts toward simultaneously maintaining motion consistency and keeping appearance consistency. The characteristics of existing prediction models in preserving spatial and temporal consistency are summarized in Table I. We find that these approaches cannot simultaneously maintain spatial and temporal consistency and thus fail to deliver satisfactory performance when applied to satellite image sequence prediction. To visually show the issue, we give prediction samples of our proposed method and state-of-the-art deep learning prediction models in Fig. 1. Notably, the satellite images are collected from the FY-4A meteorological satellite. Each frame is 1280×730 size and covers the China area. In particular, due to the GPU memory limitation, our proposed method and MSTCGAN are trained on cropped 256×256-size satellite image sequences and tested on 1280×730-size sequences to make predictions. As a lightweight prediction model, the baseline method adaptive Fourier neural operator (AFNO) is directly trained on 1280×730-size satellite image sequences to make predictions. By analyzing the prediction samples, we have the following findings here. First, AFNO makes relatively accurate predictions in terms of the locations of clouds but delivers really blurry results in terms of appearance details. Second, MSTCGAN is a GAN-based prediction method and delivers sharper prediction results, but the motion trends of clouds are quite inaccurate and unnatural. Third, compared with the baseline methods, our proposed method spatial–temporal consistency network (STCNet) more accurately predicts the motion trends of clouds and better preserves the appearance details. The illustration example shows the significance of maintaining spatial–temporal consistency. Hence, we need to address the following key obstacles to make high-quality satellite image sequence predictions.

1) How to accurately predict motion patterns of satellite image sequences to ensure temporal consistency?
2) How to enhance the appearance details of satellite image frames to guarantee spatial consistency, while the motion coherency of predicted sequences is well maintained?

To overcome the obstacles and achieve high-quality satellite image sequence prediction, we propose a novel STCNet. Specifically, it consists of two key building blocks, namely, a multilevel motion memory-based predictor and a time-variant frame discriminator. First, to accurately predict motion patterns of satellite image sequences, we carefully design a multilevel motion memory-based predictor. Notably, in the predictor, a multilevel motion memory module is proposed, which can capture and preserve motion patterns and guide the predictor to make predictions. In this manner, the predictor can effectively maintain the motion consistency of predicted sequences with ground-truth ones. Second, to enhance the spatial consistency and simultaneously keep the motion coherency of predicted sequences, we design a time-variant frame discriminator to enhance the perception quality of predicted frames. In particular, an uncertainty estimation mechanism is carefully designed, which helps the discriminator to characterize the temporal change in the predicted sequence, so as to more accurately keep the motion consistency. Third, to reduce the optimizing difficulty and enhance the training stability, we propose to apply a scheduled sampling strategy to train our proposed method STCNet. Specifically, during the training procedure, the previous predicted frame is replaced by a ground-truth frame with a certain probability to predict the next frame. Through the designed scheme, our method STCNet can be more effectively optimized and trained. To comprehensively verify the effectiveness and superiority of our method for satellite image sequence prediction, we conduct a series of experiments on satellite image sequences with different sizes (256×256, 512×512, 1280×730, and 1750×1500) from the FY-4A meteorological satellite. Besides, to further verify the applicability and adaptability of our method in more challenging real-world scenarios, we additionally conduct experiments beyond satellite image sequence prediction. In particular, we first transfer our pretrained model to predict satellite image sequences from the China area and Southeast area of Asia, respectively, to verify the spatial–temporal transfer ability of our method. Then, we conduct a convection cloud detection task on predicted sequences produced by our method to further validate the prediction accuracy of our method. All the experiment results consistently show the effectiveness of our method.

Our contributions are summarized as follows.

1) We propose a novel STCNet, which achieves high-quality satellite image sequence predictions by learning spatial–temporal consistency.
2) We propose a multilevel motion memory-based predictor. With a designed multilevel motion memory module, the predictor can more accurately predict motion patterns to ensure motion consistency.
3) The carefully designed time-variant frame discriminator recognizes and characterizes the temporal change in sequences. The discriminator enhances the appearance

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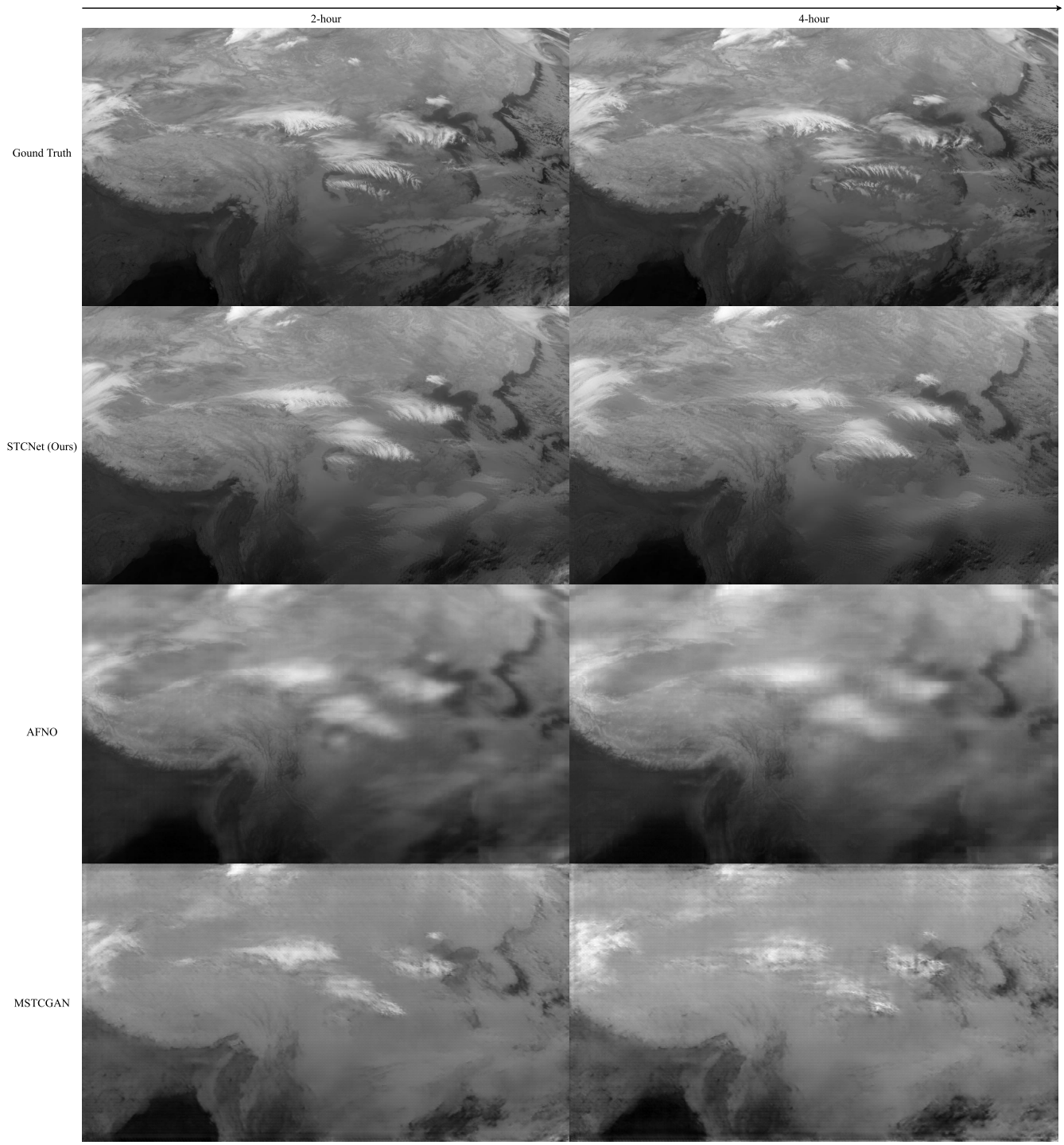

Fig. 1. 2-h (8th) and 4-h (16th) predicted frames, produced by our proposed method STCNet and the state-of-the-art deep learning prediction models. The first and second-to-fourth lines denote the ground-truth and predicted frames, respectively. Given the first eight frames, our method aims to predict the next 16 frames, and the interval between two consecutive frames is 15 min.

details of satellite image frames to keep spatial consistency and simultaneously maintains the motion coherency of the predicted sequences.

4) The designed scheduled sampling strategy can further reduce the optimizing difficulty and better train the proposed method STCNet.
5) Comprehensive experiments are conducted, which validate the effectiveness and superiority of our method for satellite image sequence prediction over state-of-the-art prediction approaches. In addition, we conduct experiments beyond satellite image sequence prediction, which further show the applicability and adaptability of our method STCNet to complex downstream tasks.

The remainder of this article is organized as follows. First, the related work is briefly introduced in Section II. Second, the proposed method STCNet is detailedly depicted in Section III.

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Third, we conduct comprehensive experiments in Section IV. Finally, the conclusion is derived in Section V.

II. Related Work

Here, we briefly introduce some related deep learning techniques.

A. Spatial–Temporal Prediction Models

Deep learning techniques [22], [23], [24], [25], [26], [27], [28], [40], [41], [42] provide valuable insights into spatial–temporal sequence prediction. ConvLSTM [40] is proposed by replacing the fully connected layers in fully-connected LSTM (FC-LSTM) [43] with convolutional layers. The model captures spatial features and temporal patterns with the convolutional layer and recurrent neural network, respectively. PredRNN [22] designs an spatio-temporal LSTM (ST-LSTM) unit, which aims to memorize both spatial appearance and temporal patterns in a unified memory pool. PhyDNet [44] proposes to disentangle physical dynamics from sequences and designs a recurrent physical cell to model physical dynamics. MotionRNN [27] designs a MotionGRU unit to capture motion patterns from the physical world and proposes a motion highway unit to avoid motion vanishing. LMC [45] provides a memory alignment learning scheme to make prediction models preserve long-term motion patterns. The aforementioned approaches mainly rely on the recurrent neural network to model temporal dynamics. Recently, some works [29], [41], [46] begin to adopt the CNN or transformer architecture to capture temporal dynamics. In particular, PredCNN [47] and SimVP [41] employ the CNN architecture to model motion patterns, which are friendly in GPU memory demand. EarthFormer [46] and Pangu [48] utilize cascaded self-attention layers to model global temporal correlation but need huge costs in terms of training data and training resources. FourCastNet [29] proposes to adopt AFNOs [49] to model spatial–temporal dynamics, which has advantages for large-scale weather forecast tasks. Although the deep learning prediction models bring up significant improvements for spatial–temporal sequence prediction, the methods make few efforts to effectively model long-term temporal patterns and deliver unsatisfactory performance in guaranteeing motion consistency for long-term prediction. In this article, we carefully design a multilevel motion memory-based predictor to effectively preserve long-term motion dynamics and more accurately make long-term satellite image sequence predictions.

B. Generative Adversarial Networks

The GAN [36] contains a generator and a discriminator. The discriminator accounts for distinguishing the real images and fake ones, while the generator aims to fool the discriminator. By conducting the adversarial training scheme, the GAN can produce high-fidelity images from Gaussian noise. However, the training procedure of the GAN is very unstable. For this, many methods [35], [37], [38], [39], [50], [51], [52] are proposed for enhancing the training stability and the quality of generated images. To alleviate the blurry issue in spatial–temporal sequence prediction, some works [2], [5], [7], [30], [31], [32], [33], [34] begin to adopt the adversarial training strategy to enhance the visual details of predicted frames. However, training a discriminator for an image sequence is really difficult, which means that the discriminator should take on the extra burden of modeling motion dynamics. For this, some works [5], [7], [30], [31], [32], [34] focus on optimizing a single predicted frame with a discriminator to enhance the perception quality and iteratively producing a whole predicted sequence. Although this manner can reduce the optimizing difficulty, it seriously destroys the motion consistency of the predicted sequence. Other works [2], [33] design specific discriminators to learn temporal dynamics. In particular, Ravuri et al. [2] designed a temporal discriminator and a spatial discriminator to capture appearance features and temporal patterns, respectively. Dai et al. [33] proposed a time-conditional discriminator to recognize predicted frames on different time steps. Although the works try to maintain the motion consistency of predicted sequences, the models cannot effectively characterize the temporal change in sequences and are still very hard to optimize. In this article, we carefully design a time-variant frame discriminator to characterize the temporal change in sequence, which can enhance the perception quality of predicted frames and simultaneously keep the motion coherency of predicted sequences. Besides, we propose a scheduled sampling strategy to reduce optimizing difficulty and better train our proposed method.

III. Proposed Method

In the following, we first give the problem definition and then detailedly introduce our proposed method STCNet.

A. Problem Definition

Here, we formally define the satellite image sequence prediction task. Assume that $X = \{x_1, x_2, \ldots, x_t\} \in \mathbb{R}^{t\times m\times n}$ and $Y = \{x_{t+1}, x_{t+2}, \ldots, x_{t+k}\} \in \mathbb{R}^{k\times m\times n}$ denote a t-frame historical sequence and a k-frame future sequence, respectively. m and n represent the length and width of each frame, respectively. Given the input sequence $X = \{x_1, x_2, \ldots, x_t\} \in \mathbb{R}^{t\times m\times n}$, our prediction model needs to generate the k-frame predicted sequence $\hat{Y} = \{\hat{x}_{t+1}, \hat{x}_{t+2}, \ldots, \hat{x}_{t+k}\} \in \mathbb{R}^{k\times m\times n}$. Hence, the task can be defined as follows:

$$\hat{Y} = \{\hat{x}_{t+1}, \hat{x}_{t+2}, \ldots, \hat{x}_{t+k}\} = \underset{x_{t+1},\ldots,x_{t+k}}{\arg\max}\, p(Y \mid X). \quad (1)$$

The predicted sequence $\hat{Y}$ is expected to be as similar as possible to the ground-truth sequence Y.

B. STCNet

Here, we introduce our proposed method STCNet in detail, which is shown in Fig. 2. Overall, STCNet contains two key building blocks, namely, a multilevel motion memory-based predictor and a time-variant frame discriminator, as shown in Fig. 2(a). First, the multilevel motion memory-based predictor accounts for predicting accurate satellite image sequences to ensure the motion consistency. Second, a time-variant frame discriminator is designed to enhance the perception quality

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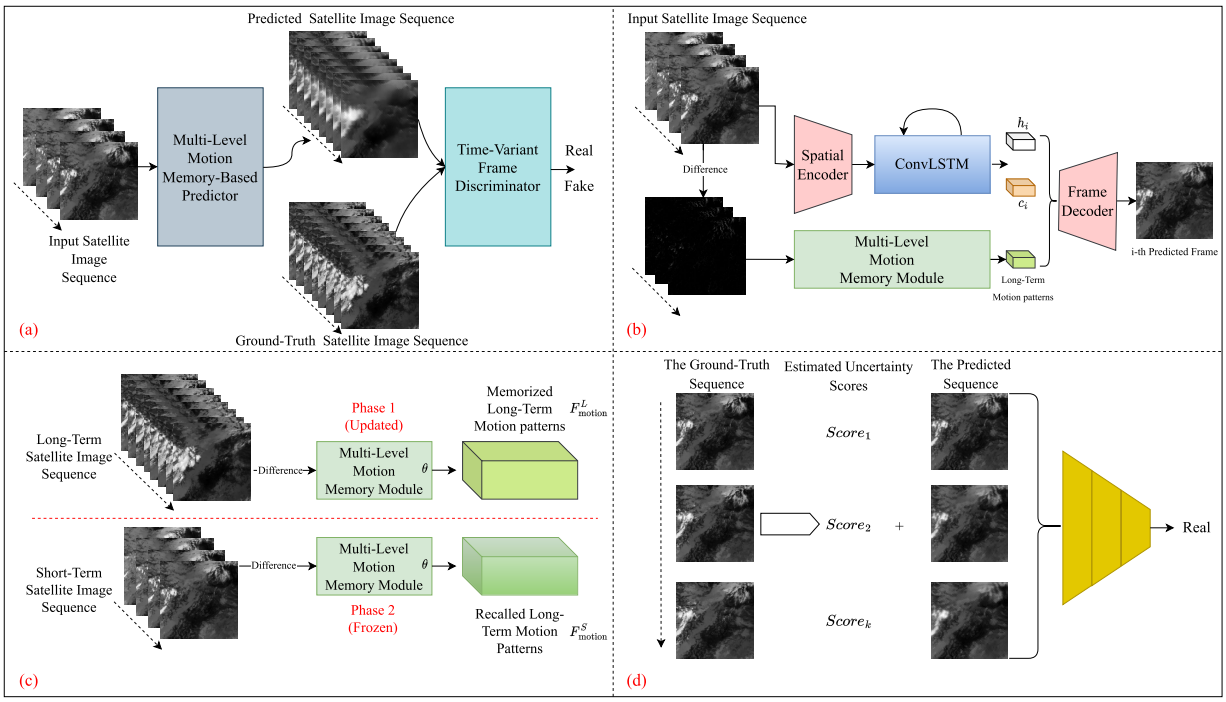

Fig. 2. Overview of our proposed method STCNet. (a) Training pipeline of STCNet. In the training procedure, the predictor and discriminator account for keeping motion and appearance consistency. Notably, in the test procedure, only the predictor works to make predictions. (b) Elements of the multilevel motion memory-based predictor. As a key part, the motion memory module aims to preserve motion patterns and guide the predictor to make predictions. (c) Two-phase training procedure of the multilevel motion memory module. In the first phase, the memory module is trained to preserve motion patterns of long-term satellite image sequences. In the test phase, short-term satellite image sequences are utilized as a query to recall possible motion patterns from the memory module. (d) Layout of the time-variant frame discriminator. With the estimated uncertainty scores, the discriminator can enhance spatial consistency and maintain motion coherency.

of predicted frames to keep spatial consistency. Notably, to simultaneously maintain the temporal coherency of the predicted sequences, an uncertainty estimating algorithm and a scheduled sampling training strategy are designed and proposed, which are beneficial to the image discriminator to characterize the temporal change in predicted sequences and enhance the training stability, respectively. Next, we elaborate on the two modules, the scheduled sampling strategy, and the model training procedure of our proposed method STCNet.

1) Multilevel Motion Memory-Based Predictor: To accurately predict motion patterns of satellite image sequences to ensure the motion consistency, we propose a multilevel motion memory-based predictor. In the predictor, we carefully design a multilevel motion memory module, which can capture and preserve multigrained motion patterns and guide the predictor to make predictions. To train our designed motion memory module, we employ the two-phase algorithm called memory alignment learning [45]. In the first phase, the memory module in the predictor is trained to store the long-term motion patterns. In the second phase, the parameters of the memory module in the predictor will be frozen, and the short-term motion dynamics is treated as a query to search for possible long-term motion patterns.

Specifically, as shown in Fig. 2(b), the predictor contains four building blocks, namely, a spatial encoder, a ConvLSTM, a multilevel motion memory module, and a frame decoder. First, the spatial encoder is utilized to encode the input satellite image sequence X into feature maps. Second, in this work, we select ConvLSTM, a classic prediction backbone model, to extract spatial–temporal dynamics from feature maps of the input sequence. Third, the designed motion memory module accounts for effectively memorizing motion patterns and guiding ConvLSTM to make predictions. Fourth, hidden state h_i and cell state c_i produced by the ConvLSTM, and long-term motion patterns produced from the motion memory module are fused by the frame decoder to generate the ith predicted frame. Next, we introduce the two-phase workflow of the predictor, which is shown in Fig. 2(c).

Assume that θ represents the memory module. In the first phase, we calculate the subtraction of consecutive frames in the long-term satellite image sequence to obtain the long-term difference sequence S^L_{motion} and then encode S^L_{motion} with our memory module θ to memorize the long-term motion patterns F^L_{motion}. The operations are formally defined as follows:

$$S^L_{\text{motion}} = Y_{2:k} - Y_{1:k-1} = \{x_{t+2-r}, x_{t+3-r}, \ldots, x_{t+k-r}\} - \{x_{t+1-r}, x_{t+2-r}, \ldots, x_{t+k-1-r}\}, \quad r \sim U\{0, t\} \tag{2}$$

$$F^L_{\text{motion}} = \theta\left(S^L_{\text{motion}}\right) \tag{3}$$

where r denotes a random offset in the long-term sequence.

The multilevel motion memory module θ captures multigrained motion patterns by multiple interactive memory branches, which is shown in Fig. 3. First, frames in the difference sequence S^L_{motion} are encoded into feature maps with a motion extractor. Second, the motion features are first

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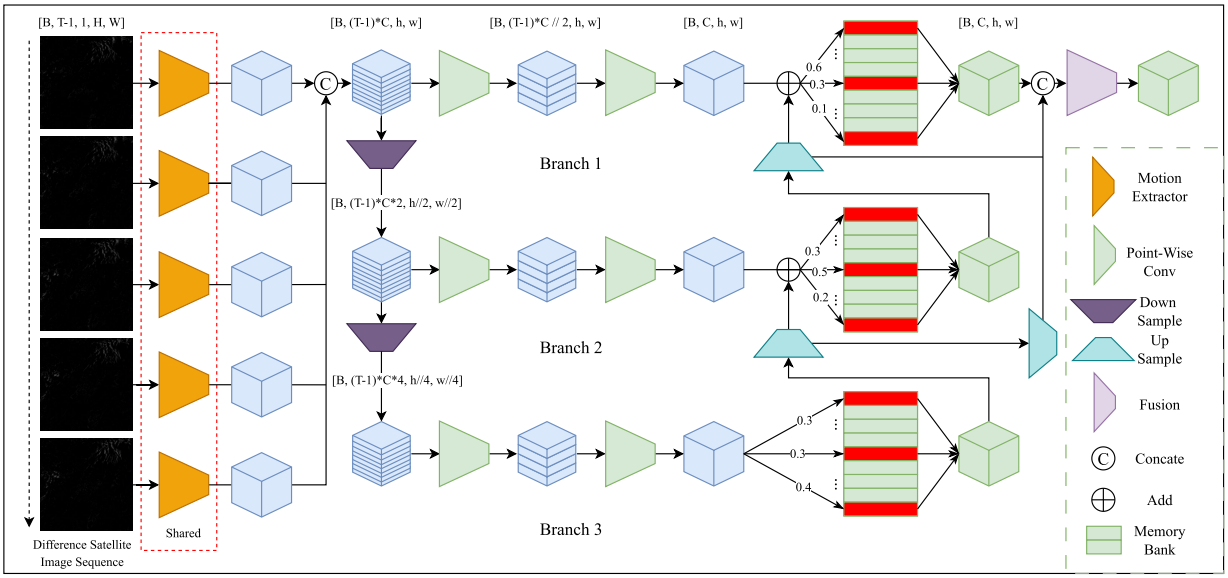

Fig. 3. Detailed structure of the designed multilevel motion memory module. The multilevel motion patterns are first captured with convolution operations and then preserved with memory banks on different branches.

concatenated along the channel direction and then downsampled to obtain higher level motion features. Third, different level motion features are further encoded with pointwise convolution operations to extract motion patterns and memorized with the corresponding memory banks. Finally, different level motion patterns are fused to obtain the multigrained motion patterns. In particular, a memory bank is actually a matrix $M \in \mathbb{R}^{N\times S}$, which can be regarded as a collection of N vectors with a size of S. Each slot $M_j (j = 1, 2, \ldots, N)$ in the memory bank can memorize one kind of possible motion pattern. The input data $z \in \mathbb{R}^S$ are first calculated similarity score with each slot M_j, and then, the output m_o of the memory bank is derived by a weighted sum of all slots. The computation process is formally defined as follows:

$$\alpha_j = \frac{\exp(\cos(z, M_j))}{\sum_{j=1}^{N} \exp(\cos(z, M_j))} \tag{4}$$

$$m_o = \frac{1}{N} \sum_{j=1}^{N} (\alpha_j \times M_j) \tag{5}$$

where $\cos(\cdot, \cdot)$ denotes the cosine similarity between two vectors.

Next, the memorized long-term motion patterns F^L_{motion} and hidden states h_i and c_i produced by ConvLSTM are fused to produce the ith predicted frame $\hat{Y}_i$. F^L_{motion} represents motion patterns of a long-term sequence and should be refined to adapt the ith frame prediction. Since c_i contains motion dynamics at the ith time step, we use c_i to refine F^L_{motion}. Specifically, c_i and F^L_{motion} are first concatenated and projected into channelwise attention score β_i with full-connected layers. Then, we can obtain the refined motion patterns $\hat{F}^L_{\text{motion}} = F^L_{\text{motion}} \otimes \beta_i$ for the ith frame prediction. Finally, $\hat{F}^L_{\text{motion}}$ and h_i are first concatenated and then decoded by the frame decoder into ith predicted frame.

In the second phase, we utilize the short-term difference sequence S^S_{motion} to recall possible long-term motion patterns F^S_{motion} from our motion memory module θ to guide frame prediction. Formally, the operations are defined as follows:

$$S^S_{\text{motion}} = X_{2:t} - X_{1:t-1} = \{x_2, x_3, \ldots, x_t\} - \{x_1, x_2, \ldots, x_{t-1}\} \tag{6}$$

$$F^S_{\text{motion}} = \theta\left(S^S_{\text{motion}}\right). \tag{7}$$

The other operations for producing the ith frame are the same as phase 1. By conducting the two-phase training algorithm, the predictor can more accurately predict the motion patterns of satellite image sequences with guidance of the motion memory module. Notably, in the test procedure, the predictor only conducts phase 2 to make satellite image sequence predictions.

2) Time-Variant Frame Discriminator: As shown in Fig. 2(d), to enhance the perception quality of the predicted frames produced by the predictor and simultaneously maintain the motion coherency of the predicted sequences, we carefully design the time-variant frame discriminator by addressing a key problem of effectively training our method, that is, how to accurately balance the GAN loss for image quality and mse or MAE loss for prediction accuracy in terms of prediction time step. As the prediction time step increases, the divergence between the ground-truth frame and the predicted frame is growing. However, the original discriminator in GANs equally treats the frames in different time steps, which fails to recognize the temporal change and tends to destroy the motion consistency of the predicted sequences. Actually, as the prediction time step increases, the prediction uncertainty dramatically increases. Correspondingly, the function of GAN loss on the predicted frames should fade as the prediction time increases, compared to the mse or MAE loss. With this intuition, we design an uncertainty estimation mechanism to help the discriminator to quantify the temporal change. The mechanism estimates the uncertainty of future satellite image sequences by calculating the appearance discrepancy

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Algorithm 1 Mini-Batch Estimating Uncertainty

Input: The mini-batch past satellite image sequence $X = \{x_1, x_2, \ldots, x_t\}$; the mini-batch future satellite image sequence $Y = \{x_{t+1}, x_{t+2}, \ldots, x_{t+k}\}$.

Output: The estimated scores $\text{score} = \{\text{score}_1, \text{score}_2, \ldots, \text{score}_k\}$.

Initialize: The pretrained model FlowNet 2.0; the hyperparameter λ; $\text{motion}_0 = 0$.
foreach $i = 1, 2, \ldots, k$ **do**
  $\text{appearance}_i = \text{sum}(|x_t - x_{t+i}|)$;
  $u, v = \text{FlowNet2.0}(x_{t+i-1}, x_{t+i})$;
  $d = \max(\sqrt{u^2 + v^2})$;
  $\text{motion}_i = \text{sum}(\sqrt{(u/d)^2 + (v/d)^2}) + \text{motion}_{i-1}$;
end
foreach $i = 1, 2, \ldots, k$ **do**
  $\text{appearance}_i = \text{appearance}_i / \text{appearance}_1$;
  $\text{motion}_i = \text{motion}_i / \text{motion}_1$;
  $\text{score}_i = (0.5 * \text{appearance}_i + 0.5 * \text{motion}_i)^{\lambda}$;
end

and motion distance between the two consecutive frames. Specifically, we utilize the MAE loss and an opticalflow model FlowNet 2.0 [19] to compute the appearance and motion uncertainty of satellite image sequences, respectively. The estimating process is formally defined in Algorithm 1, where u and v denote the horizontal and vertical offset matrices, respectively. The hyperparameter λ serves as a power exponent to scale up or down the scores. The estimated scores will be utilized to distinguish the impact of GAN on predicted frames at different time steps, which will be detailedly introduced in Section III-B4. Moreover, to enhance the training stability, we propose to utilize the spectral normalization [39] to restrict the weights of each convolutional layer, which ensures the discriminator to satisfy the Lipschitz constraint.

3) Scheduled Sampling: Although we design the time-variant frame discriminator, directly optimizing the whole predicted sequence with the GAN is still hard. To further reduce the training difficulty, we propose to apply the scheduled sampling strategy to train our method. In the normal prediction manner, the previous predicted frame is treated as the input to generate the next predicted frame, which causes the accumulation error and also makes it difficult to optimize. With the scheduled sampling mechanism, the previous predicted frame is replaced by a ground-truth frame with a certain probability p to predict the next frame, which is defined as follows:

$$\hat{Y}_i(i > t+1) = \begin{cases} P\left(\hat{Y}_{i-1}, F^S_{\text{motion}}\right), & (1-p) \\ P\left(Y_{i-1}, F^S_{\text{motion}}\right), & (p) \end{cases} \tag{8}$$

where $P(.)$ denotes the predictor. Specifically, we give the curve of sampling possibility with respect to training iteration in Fig. 4. First, we utilize a ground-truth frame with a fixed possibility to predict the next frame. Then, we gradually decrease the sampling possibility. Finally, we totally rely on the predicted frame to predict the next frame. In this

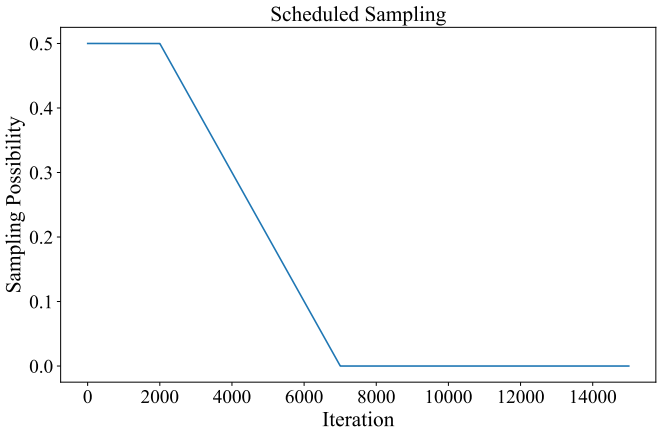

Fig. 4. Curve of sampling possibility with respect to training iteration numbers.

progressive manner, the predictor and discriminator in GANs can be effectively coordinated and our method can be more effectively trained.

4) Model Training Procedure: Here, we summarize the training procedure of our model, which consists of two stages. In the first stage, the time-variant frame discriminator is trained to distinguish the predicted frame and ground-truth frame, while the parameters of the multilevel motion memory-based predictor are frozen. In the second stage, the predictor is trained to make high-quality predictions, while the parameters of the discriminator are frozen. Next, we introduce the two stages in detail.

1) *Discriminator Training:* The time-variant frame discriminator D aims to classify the ground-truth observation frame Y_i into label 1 and the prediction frame $\hat{Y}_i$ into label 0. Thus, the loss function for D is

$$L_D(\hat{Y}, Y) = \sum_{i=1}^{k} \{L_{\text{BCE}}(\hat{Y}_i, 0) + L_{\text{BCE}}(Y_i, 1)\} \tag{9}$$

$$L_{\text{BCE}}(\hat{y}, y) = -(y \ln(\hat{y}) + (1-y)\ln(1-\hat{y})) \tag{10}$$

where L_{BCE} is the binary cross-entropy function and $\hat{y}$ and y represent the prediction and the label, respectively. Notably, (9) is optimized with the scheduled sampling strategy.

2) *Predictor Training:* As aforementioned in Section III-B1, the predictor contains two phases for training. In the first phase, the multilevel motion memory-based predictor is trained to memorize the long-term satellite image motion patterns. We utilize the combination of MAE and mse losses to train the model, which is defined as follows:

$$L(\hat{Y}, Y) = \sum_{i=1}^{k} \left|\hat{Y}_i - Y_i\right| + \sum_{i=1}^{k} \left|\hat{Y}_i - Y_i\right|^2. \tag{11}$$

In the second phase, the predictor is trained to make long-term satellite image sequence predictions. In particular, we design a combined loss L_G for the multilevel motion memory-based predictor, which consists of four losses, namely, an image discriminative loss L_D, a time-variant MAE

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loss L_{TMAE}, a time-variant perceptual loss L_{TP}, and a total variation loss L_{TV}.

First, the image discriminative loss L_D is utilized to fool the time-variant frame discriminator, which is defined as follows:

$$L_D(\hat{Y}) = \sum_{i=1}^{k} \{L_{\mathrm{BCE}}(\hat{Y}_i), 1)\}. \tag{12}$$

Second, the time-variant MAE loss L_{TMAE} is utilized to constrain the predictor to fit ground-truth sequences. As we have introduced in Section III-B2, the estimated uncertainty scores will be utilized to differentiate the impact of GAN on different predicted frames. To this end, we apply the scores to the MAE loss, and the time-variant MAE loss is formally defined as follows:

$$L_{\mathrm{TMAE}}(\hat{Y}, Y) = \sum_{i=1}^{k} \mathrm{score}_i \times |\hat{Y}_i - Y_i|. \tag{13}$$

Through (13), the impact of MAE loss on predicted frames strengthens as the prediction time step increases, and as a corresponding result, the impact of the GAN on predicted frames gradually decreases. In this manner, our method STCNet can be more effectively trained.

Third, different from L_{TMAE}, the time-variant perceptual loss L_{TP} measures the discrepancy between the predicted sequence and the ground-truth sequence on the high level, which benefits preserving the fine-grained texture details. Besides, as the prediction time step increases, the weight of the loss on the predicted frame should decrease to avoid unnecessary noises. Hence, we apply the reciprocal of estimated uncertainty scores to the perceptual loss. The time-variant perceptual loss L_{TP} is defined as follows:

$$\begin{aligned} L_{\mathrm{TP}}(\hat{Y}, Y) = &\frac{1}{8} * L^1_{\mathrm{TP}}(\hat{Y}, Y) \\ &+ \frac{1}{4} * L^2_{\mathrm{TP}}(\hat{Y}, Y) \\ &+ \frac{1}{2} * L^3_{\mathrm{TP}}(\hat{Y}, Y) \\ &+ L^4_{\mathrm{TP}}(\hat{Y}, Y) \end{aligned} \tag{14}$$

$$L^j_{\mathrm{TP}}(\hat{Y}, Y) = \sum_{i=1}^{k} \frac{1}{\mathrm{score}_i} \times \frac{1}{C_j * H_j * W_j} |\phi_j(\hat{Y}_i) - \phi_j(Y_i)|^2 \tag{15}$$

where $\phi_j(x)$ denotes a feature of shape $C_j * H_j * W_j$ produced by the jth convolutional layer in a pretrained model VGG for an input image x and L_{TP} is a weighted combination of the first four convolutional layers in VGG. Through the designed L_{TP}, the motion coherency of the predicted sequence can be carefully kept.

Fourth, the total variation loss L_{TV} is employed to guarantee the appearance smoothness in the prediction frame, which is defined as

$$\begin{aligned} L_{\mathrm{TV}}(\hat{Y}) = \sum_{i=1}^{k} \sum_{r=1}^{m} \sum_{s=1}^{n} &|\hat{Y}_i(r, s+1) - \hat{Y}_i(r, s)| \\ &+ |\hat{Y}_i(r+1, s) - \hat{Y}_i(r, s)| \end{aligned} \tag{16}$$

where $\hat{Y}_i(r, s)$ denotes the pixel at the rth row and sth column from the ith prediction frame $\hat{Y}_i$.

Finally, the combined loss L_G is formally defined as follows:

$$\begin{aligned} L_G(\hat{Y}, Y) = &\lambda_1 * L_{\mathrm{TMAE}}(\hat{Y}, Y) + \lambda_2 * L_{\mathrm{TP}}(\hat{Y}, Y) \\ &+ \lambda_3 * L_{\mathrm{TV}}(\hat{Y}) + L_D(\hat{Y}) \end{aligned} \tag{17}$$

where the hyperparameters λ_1–λ_3 are corresponding weights for L_{TMAE}, L_{TP}, and L_{TV}, respectively. Similarly, (17) is optimized with the scheduled sampling strategy. For clarity, the detailed training procedure of our method STCNet is summarized as in Algorithm 2. We note that in the training procedure of our memory-based predictor, phases 1 and 2 will be alternatively conducted, while in the test procedure, only phase 2 will be used to make the long-term satellite image sequence prediction.

Algorithm 2 Training Algorithm of the Proposed Method STCNet

Input: The t-frame historical satellite image sequence $X = \{x_1, x_2, \ldots, x_t\}$; k-frame future satellite image sequence $Y = \{x_{t+1}, x_{t+2}, \ldots, x_{t+k}\}$.

Setting: Set the learning rate l for the proposed model; Set λ_1, λ_2, and λ_3 for L_G; Set d for the discriminator iteration.
while not converged **do**
  Sample the training batch of input sequences;
  Discriminator Training:
  Froze parameters of the multi-level motion memory-based predictor G.
  foreach $i = 1, 2, \ldots, d$ **do**
    Update parameters of the time-variant frame discriminator D:
    $D = D - l * \frac{\partial L_D(\hat{Y}, Y)}{\partial D}$, and $L_D(\hat{Y}, Y)$ is computed as formulae (4)-(10).
  end
  Predictor Training:
  Froze parameters of the time-variant frame discriminator D.
  if *phase 1* **then**
    Update the parameters of the predictor G:
    $G = G - l * \frac{\partial (L(\hat{Y}, Y)}{\partial G}$, and $L(\hat{Y}, Y)$ is computed as formulae (2)-(5) (8), and (11).
  end
  if *phase 2* **then**
    Froze the parameters of the memory bank θ in the predictor and update other parameters:
    $G = G - l * \frac{\partial (L(\hat{Y}, Y)}{\partial G}$, and $L(\hat{Y}, Y)$ is computed as formulae (4)-(8) and (12)-(17).
  end
end

IV. Experiments

In this section, we first conduct comprehensive experiments to validate the effectiveness and superiority of our proposed

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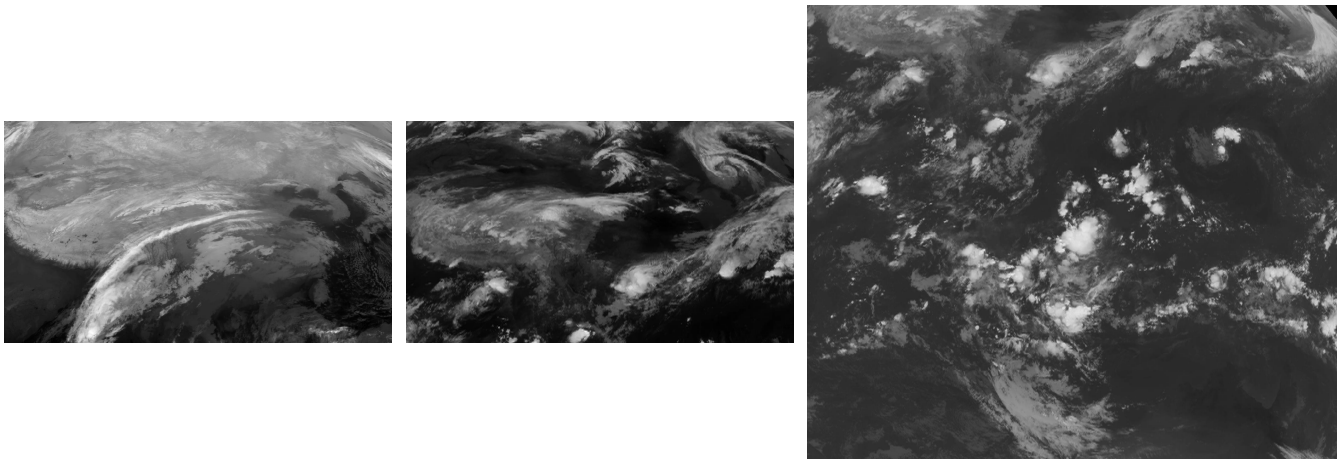

(a) (b) (c)

Fig. 5. Satellite image samples at different times and regions from the FY-4A meteorological satellite. (a) Satellite image from China area at 23:30 on January 1, 2018. (b) Satellite image from China area at 00:00 on June 1, 2020. (c) Satellite image from the Southeast Asia area at 00:00 on June 1, 2020.

method STCNet. Second, detailed ablation experiments are conducted to verify the effectiveness of key components in our proposed method. Third, we apply our method to other downstream tasks beyond satellite image sequence prediction, which further shows the applicability and adaptability of our method in real-world scenarios.

A. Experimental Setup

1) FY-4A Satellite Cloud Image Datasets: FY-4A is a meteorological satellite launched by China in 2016, which can generate satellite images in 14 different wavebands with a multichannel visible infrared scanning imager. Specifically, the 14 channels can be categorized into five types, namely, visible and near-infrared (1–3), short-wave infrared (4–6), midwave infrared (7 and 8), water vapor (9 and 10), and long-wave infrared (11–14). Among the 14 channels, the 12th channel is suitable for observing the moving trajectory of clouds in the atmosphere, which is significant for convection weather nowcast. In this work, we construct three extremely challenging datasets from the FY-4A meteorological satellite images at the 12th channel. The first dataset consists of 1280×730-size satellite image sequences from the China area in 2018, while the second one contains sequences from the China area in June 2020. Different from the previous two datasets, the third dataset contains 1750×1500-size satellite image sequences from the Southeast Asia area in June 2020. Here, we give satellite image samples of the three datasets in Fig. 5. The brief descriptions of the three datasets are given as follows.

1) *Dataset 1:* It is constructed on satellite images from the FY-4A in 2018. Specifically, it contains 11 669 satellite image sequences in the China area with a length of 24, where each pixel denotes a 4×4 km region. The interval between the two consecutive images is 15 min. We utilize the first 8-frame (2-h) sequences as input and aim to predict the next 16-frame (4-h) sequences. Besides, the training set, validation set, and test set are constructed according to ratios of 70%, 15%, and 15%, respectively. Notably, during the training procedure, each frame in sequences is obtained by cropping the 256×256 parts from the original 1280×730-size satellite images, due to the GPU memory limitation. However, since our proposed method is a fully convolutional network, once trained, it can be tested with arbitrary sizes, such as 256×256 and 512×512.
2) *Dataset 2:* Different from dataset 1, it consists of 1374 satellite image sequences in June 2020. Since the atmospheric motion is fierce in June (summer), sequences in the dataset are more difficult to predict.
3) *Dataset 3:* The dataset contains 1990 satellite image sequences in June 2020 from the Southeast Asia area, which is another region different from the China area in datasets 1 and 2. The size of each frame is 1750×1500. Due to the scanning mechanism of FY-4A, in the area, the interval between two consecutive frames is irregular, and we interpolate frames to keep the interval that is 15 min.

2) Evaluation Metrics: In this work, we utilize five metrics for spatial–temporal sequence prediction to evaluate the prediction performance of our method, including mse [33], MAE [33], peak-signal-to-noise ratio (PSNR) [33], gradient difference loss (GDL) [33], and learned perceptual image patch similarity (LPIPS) [53]. The higher PSNR score denotes the better pixelwise quality of the predicted sequence, while the lower mse and MAE scores represent the lower prediction error between the ground-truth sequence and predicted sequence. The GDL score measures the appearance sharpness similarity between the ground-truth sequence and predicted sequence, and the lower score denotes a better prediction quality. Different from the previous four metrics, LPIPS is a high-level evaluation metric, and the lower LPIPS score denotes better perceptual quality. In a word, the higher PSNR score the better, while the lower mse, MAE, GDL, and LPIPS scores the better. Notably, the PSNR, mse, MAE, and GDL scores are calculated on standard satellite images, whose gray values belong to [0, 255]. The LPIPS score is computed based on scaled images, whose gray values are normalized to [0, 1].

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TABLE II
DETAILED PARAMETERS OF THE TIME-VARIANT FRAME DISCRIMINATOR

Time-Variant Frame Discriminator			
Layer	Output Channels	Stride	Kernel Size
Conv2d	64	1	3 × 3
Conv2d	64	2	4 × 4
Conv2d	128	1	3 × 3
Conv2d	128	2	4 × 4
Conv2d	256	1	3 × 3
Conv2d	256	2	4 × 4
Conv2d	512	1	3 × 3

3) Environment: In this article, all the experiments are conducted on a GPU server with a CPU i7-9700K at 3.60 GHz, a 64-GB memory, and a NVIDIA RTX8000.

4) Implementation Details: Our method STCNet is trained by an Adam optimizer [54] with a learning rate of 0.0002. The channel of feature maps produced by the spatial encoder and motion extractor is 128. As for the multilevel motion memory-based predictor, the shape of the three memory banks, short length, long length, and the depth of ConvLSTM, is set to 400 × 128, 200 × 256, and 100 × 512, and 8, 16, and 4, respectively. λ in Algorithm 1 is set to 0.6. As for the combined loss L_G, λ_1–λ_3 are set to 300, 1200, and 600, respectively. Specifically, the discriminator iteration is set to 1, and the detailed parameters of the predicted frame discriminator are shown in Table II, where each Conv2d layer is followed by a leakyReLU [55] activation function with the leak value of 0.1.

5) Baseline Methods: We compare our method STCNet with 15 baseline methods, including Opticalflow-LK [18], ConvGRU [26], ConvLSTM [40], TrajGRU [56], PredRNN [22], PredRNN++ [23], MIM [24], PhyDNet [44], SA-ConvLSTM [25], Conv-TT-LSTM [57], MotionRNN [27], LMC [45], SimVP [41], MSTCGAN [33], and AFNO [49]. Among the approaches, Conv-TT-LSTM, MotionRNN, LMC, SimVP, MSTCGAN, and AFNO are the newest state-of-the-art methods.

B. Results on Long-Term Satellite Image Sequence Prediction

1) Quantitative Results Comparison: To verify the effectiveness of our proposed method on satellite image sequences with different sizes, we conduct experiments on 256 × 256-size and 512 × 512 -size satellite image sequences. According to the workflow of the baseline methods, the 15 approaches are categorized into two types, namely, the opticalflow-based method and deep learning prediction methods. In particular, four deep learning models, including TrajGRU, PhyDNet, MotionRNN, and AFNO, are not size-free prediction models, which means that the methods can only be tested with 256 × 256-size satellite image sequences. In this work, we report the quantitative results of the three models on 512 × 512-size satellite image sequences in a simple manner. Specifically, we first divide the 512 × 512-size image into four no-overlapped patches to make predictions and then concatenate them into a predicted frame.

Table III shows the quantitative prediction results of the 15 baseline methods and our proposed method STCNet. Obviously, our method STCNet delivers the best performance both on the 256 × 256-size and 512 × 512 -size evaluation tasks in terms of the PSNR, mse, MAE, and LPIPS metrics. First, our method STCNet outperforms SimVP (the second-best method) by 1.54% in PSNR, 5.57% in mse, and 7.43% in MAE, and improves Opticalflow-LK by 6.40% in LPIPS on the 256 × 256-size evaluation task. Second, our method STCNet improves LMC by 1.90% in PSNR, 10.16% in mse, and 9.19% in MAE, and outperforms Opticalflow-LK by 1.78% in LPIPS on the 512 × 512-size evaluation task. Third, our method STCNet outperforms LMC by 1.85% in PSNR, 13.46% in mse, and 9.96% in MAE, and improves Opticalflow-LK by 4.09% in LPIPS on the average evaluation. The observations demonstrate the effectiveness and advantage of our method over state-of-the-art methods for satellite image sequence prediction. Moreover, the superiority of our method becomes more obvious when producing larger size satellite image sequences.

As an opticalflow-based prediction approach in the baseline methods, Opticalflow-LK delivers unsatisfactory performance in terms of the PSNR, mse, MAE, and GDL scores but significantly outperforms other deep learning models in terms of LPIPS score. The observation is reasonable since producing sharp predicted frames is a natural advantage of the opticalflow-based prediction method. However, precisely estimating the motion field of satellite image sequences is extremely difficult. As a result, the prediction results of the method are quite inaccurate, which indicates the limitation of the opticalflow method on long-term satellite image sequence prediction. Different from the opticalflow method, deep learning prediction methods deliver competitive performance in terms of PSNR, mse, MAE, and GDL scores. In particular, SimVP achieves the second-best performance in terms of the PSNR, mse, and MAE metrics on 256 × 256-size satellite image sequences evaluation task. LMC delivers the second-best performance in terms of the PSNR, mse, and MAE scores on 512 × 512-size satellite image sequences evaluation task. Besides, Conv-TT-LSTM and MotionRNN, and MSTCGAN show their advantages on the GDL and LPIPS scores, respectively. Compared with the baseline methods, our proposed method STCNet consistently delivers the best or second-best performance on the five metrics, which shows that our method can simultaneously ensure the accuracy and sharpness of prediction results.

To investigate performance with respect to lead time, we depict PSNR, mse, MAE, GDL, and LPIPS curves in Fig. 6. As lead time increases, the performance of each method degrades. It is easy to understand since prediction becomes more challenging as the lead time increases. Among the methods, our approach delivers the best performance in terms of the PSNR, mse, MAE, and LPIPS scores, and the advantage becomes more obvious, especially in the last 2 h. Although the proposed STCNet performs slightly worse than Conv-TT-LSTM and MotionRNN in terms of the GDL score, our method can make much more accurate predictions in terms of the PSNR, mse, MAE, and LPIPS scores.

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TABLE III

QUANTITATIVE PREDICTION RESULTS OF BASELINE METHODS AND STCNET ON 256 × 256-SIZE AND 512 × 512-SIZE SATELLITE IMAGE SEQUENCES, RESPECTIVELY. RED DENOTES THE BEST AND BLUE DENOTES THE SECOND BEST

Model	Size Free	256 × 256					512 × 512					Average				
		↑ PSNR	↓ MSE	↓ MAE	↓ GDL	↓ LPIPS	PSNR	MSE	MAE	GDL	LPIPS	PSNR	MSE	MAE	GDL	LPIPS
Opticalflow-LK [18]	Yes	24.40	384.62	12.58	4.98	**0.172**	24.13	404.78	12.85	4.90	**0.169**	24.27	394.70	12.72	4.94	**0.171**
ConvGRU [26]	Yes	24.48	351.85	13.02	4.65	0.352	24.06	371.88	13.41	4.50	0.357	24.27	361.87	13.22	4.58	0.355
ConvLSTM [40]	Yes	25.00	324.39	12.64	4.78	0.384	24.28	388.63	13.98	4.64	0.396	24.64	356.51	13.31	4.71	0.390
TrajGRU [56]	No	25.01	341.37	12.93	4.72	0.357	23.26	604.79	16.10	5.73	0.381	24.14	473.08	14.52	5.23	0.369
PredRNN [22]	Yes	25.06	337.12	12.84	4.86	0.434	24.12	410.62	14.67	4.64	0.412	24.59	373.87	13.75	4.75	0.423
PredRNN++ [23]	Yes	24.75	337.38	13.33	4.84	0.406	24.30	369.76	14.09	4.69	0.411	24.53	353.57	13.71	4.77	0.409
MIM [24]	Yes	24.67	329.35	13.02	4.84	0.405	24.10	375.16	14.18	4.70	0.413	24.39	352.26	13.60	4.77	0.409
PhyDNet [44]	No	24.21	327.41	13.08	4.86	0.461	23.26	388.52	13.88	4.93	0.463	23.74	357.97	13.48	4.90	0.462
SA-ConvLSTM [25]	Yes	24.08	366.74	13.99	13.63	0.449	23.31	449.29	15.72	13.74	0.452	23.70	408.02	14.86	13.69	0.451
Conv-TT-LSTM [57]	Yes	25.07	430.53	13.39	**4.58**	0.293	24.87	434.12	13.12	**4.43**	0.299	24.97	432.33	13.26	**4.51**	0.296
MotionRNN [27]	No	24.48	495.41	13.83	**4.59**	0.350	23.21	632.43	16.12	4.64	0.379	23.85	563.92	14.98	4.62	0.365
LMC [45]	Yes	26.52	234.15	9.76	4.96	0.446	**26.31**	**243.18**	**9.79**	4.82	0.457	**26.42**	**238.67**	**9.78**	4.89	0.452
SimVP [41]	Yes	**26.59**	**223.77**	**9.69**	4.80	0.447	25.90	269.69	10.83	4.65	0.450	26.25	246.73	10.26	4.73	0.449
MSTCGAN [33]	Yes	25.41	280.91	11.06	4.68	0.233	24.33	326.25	12.96	4.53	0.262	24.87	303.58	12.01	4.61	0.248
AFNO [49]	No	25.22	271.63	11.32	8.25	0.431	24.46	331.59	12.52	8.58	0.470	24.84	301.61	11.92	8.42	0.451
STCNet (Ours)	Yes	**27.00**	**211.31**	**8.97**	4.61	**0.161**	**26.81**	**218.47**	**8.89**	**4.47**	**0.166**	**26.91**	**214.89**	**8.93**	**4.54**	**0.164**

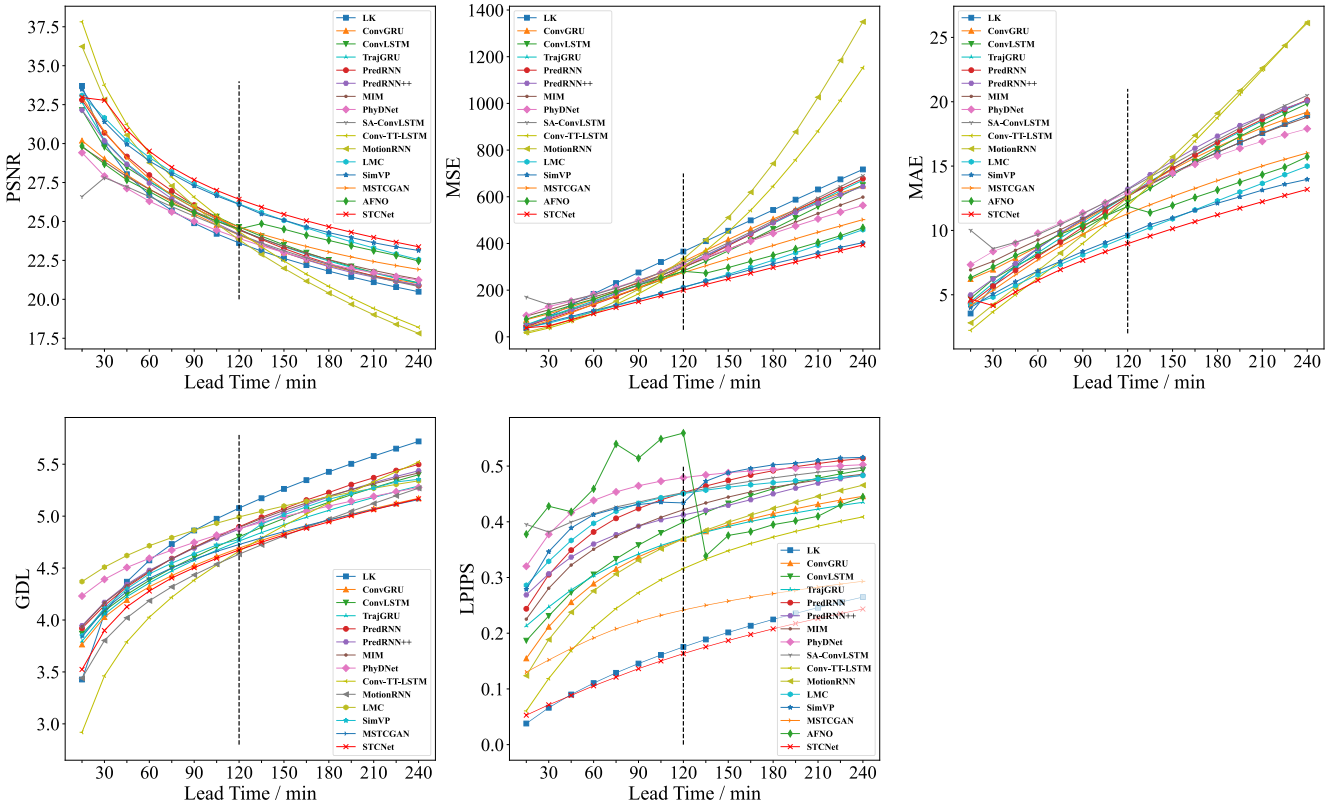

Fig. 6. PSNR, mse, MAE, GDL, and LPIPS curves of baseline methods and STCNet on 256 × 256-size satellite image sequences. In particular, the GDL curves of SA-ConvLSTM and AFNO are not included for their relatively bad performance.

To compare the experimental cost of the baseline methods and our proposed method STCNet, we give quantitative cost results in Table IV. Among the prediction models, our approach STCNet shows competitive performance in terms

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TABLE IV

QUANTITATIVE MODEL COST OF BASELINE METHODS AND STCNET ON 256 × 256-SIZE SATELLITE IMAGE SEQUENCES. THE GPU MEMORY AND INFERENCE TIME DENOTES THE COST OF PER SATELLITE IMAGE SEQUENCE FOR TRAINING AND TESTING, RESPECTIVELY. THE MODELS ARE EVALUATED ON AN RTX 8000 GPU WITH 48-GB MEMORY

Prediction Method	Performance 8 → 16			
	FLOPs	GPU Memory	Inference Time	#param
Conv-TT-LSTM	2155G	40.9G	1.206s	2.69M
MotionRNN	2274G	9.71G	3.196s	91.04M
LMC	809G	8.81G	0.103s	33.91M
SimVP	123G	6.33G	0.038s	15.34M
MSTCGAN	1911G	12.30G	0.569s	23.08M
AFNO	31G	3.01G	0.029s	61.86M
STCNet (ours)	679G	14.97G	0.166s	61.63M

of floating point operations (FLOPs), inference time, and parameters. Though the baseline methods SimVP and AFNO are efficient in terms of computation, they deliver very blurry prediction results and fail to achieve high-quality predictions.

2) Qualitative Results Comparison: To visually compare the prediction quality, we give 256 × 256-size and 512 × 512 -size prediction samples of typical baseline methods and our proposed method STCNet in Figs. 7 and 8, respectively. By comparing the prediction samples, we have the following findings here. First, though Conv-TT-LSTM and MotionRNN produce pretty sharp frames, the appearance details and cloud locations are far from the ground truth. Besides, the size of predicted frames from MotionRNN is dependent on the training sequences, which further obstacles to its applicability in real-world scenarios. Second, LMC and SimVP make more accurate predictions in terms of the cloud locations yet much more blurry in terms of appearance details. Third, as a GAN-based prediction method, MSTCGAN produces accurate frames in terms of appearance details, but the motion patterns of the predicted sequences are incoherent. Specifically, in the last 2 h, the moving trajectory of the clouds is far from the ground truth. Fourth, AFNO delivers very blurry prediction results and it is also not a fully convolutional prediction method, which limits its potential for producing larger size satellite image sequences. Finally, our proposed method STCNet makes better predictions both in terms of appearance details and motion patterns. The observations again demonstrate the effectiveness and superiority of our method on long-term satellite image sequence prediction.

C. Ablation Study

To verify the effectiveness of key building blocks and our carefully designed structures in STCNet, we obtain five variants by ablating our proposed method as follows.

1) *Variant-1:* STCNet w/o time-variant frame discriminator. The variant is obtained by directly removing the time-variant frame discriminator in the proposed STCNet.
2) *Variant-2:* Based on variant-1, removing downsample and upsample operations in the multilevel motion memory module, and shapes of memory banks in the three branches are set to 400 × 128, 400 × 128, and 400 × 128.
3) *Variant-3:* Based on variant-1, the second and third motion memory branches are removed.
4) *Variant-4:* STCNet w/o uncertainty estimation mechanism. Based on variant-1, an original discriminator is added but without the uncertainty estimation mechanism.
5) *Variant-5:* STCNet w/o scheduled sampling. Based on variant-4, the uncertainty estimation mechanism is added. Compared with the proposed STCNet, the variant is trained without the scheduled sampling strategy.

TABLE V

QUANTITATIVE PREDICTION RESULTS OF STCNET AND ITS VARIANT METHODS ON 256 × 256-SIZE SATELLITE IMAGE SEQUENCES. RED DENOTES THE BEST AND BLUE DENOTES THE SECOND BEST

Prediction Method	Performance 8 → 16				
	PSNR ↑	MSE ↓	MAE ↓	GDL ↓	LPIPS ↓
Variant-1	**27.15**	**205.51**	**8.93**	4.78	0.420
Variant-2	**27.02**	**207.60**	9.15	4.82	0.433
Variant-3	26.76	216.24	9.32	4.79	0.428
Variant-4	26.64	228.34	9.45	**4.71**	0.206
Variant-5	26.90	212.49	9.09	4.82	**0.205**
STCNet	27.00	211.31	**8.97**	**4.61**	**0.161**

Table V shows the quantitative prediction results of STCNet and its variant methods. Obviously, our proposed method STCNet delivers the best performance on the GDL and LPIPS scores and the second or third best performance on the MAE, PSNR, and mse scores, which is very competitive to the best corresponding scores. The observation indicates that our approach can effectively enhance the appearance details of predicted frames while simultaneously maintaining the prediction accuracy as much as possible. In particular, by analyzing the ablation experiment results, we obtain six conclusions here. First, variant-1 delivers the best performance in terms of the three low-level metrics, including PSNR, MAE, and mse, but performs relatively badly in terms of the GDL and LPIPS scores. This observation suggests that, without the function of a discriminator, the multilevel motion memory-based predictor is able to make accurate predictions, yet inevitably loses appearance details and produces blurry predicted frames. Second, variant-1 consistently outperforms variant-2 on the five metrics, which demonstrates that downsample operations in the motion memory module are beneficial for capturing higher level motion patterns. Third, variant-1 significantly outperforms variant-3 in terms of PSNR, mse, and MAE scores. This observation shows the significance of our designed multilevel structure for effectively capturing motion patterns. Fourth, variant-4 outperforms variant-1 in terms of the GDL and LPIPS scores but delivers unsatisfactory performance in terms of the PSNR, mse, and MAE scores. The observation indicates that the original discriminator is beneficial to enhance the perception quality of predicted frames but fails to model the predicted sequences at the sequence level and tends to produce inaccurate predictions.

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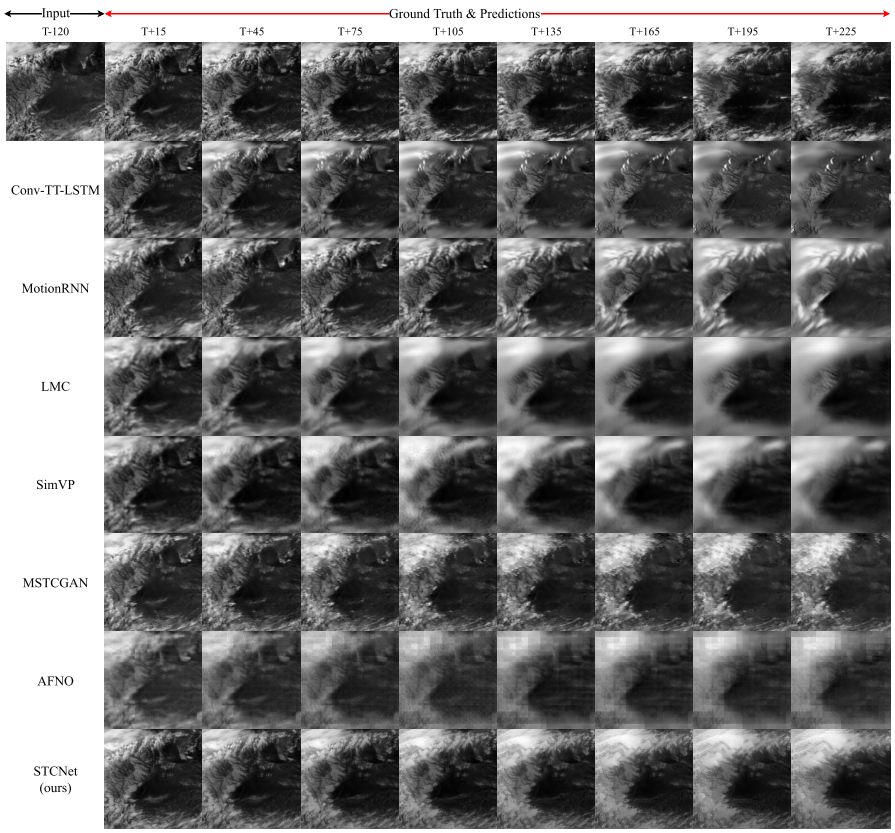

Fig. 7. 256 × 256-size predictions of baseline methods and STCNet. Every two frames are shown.

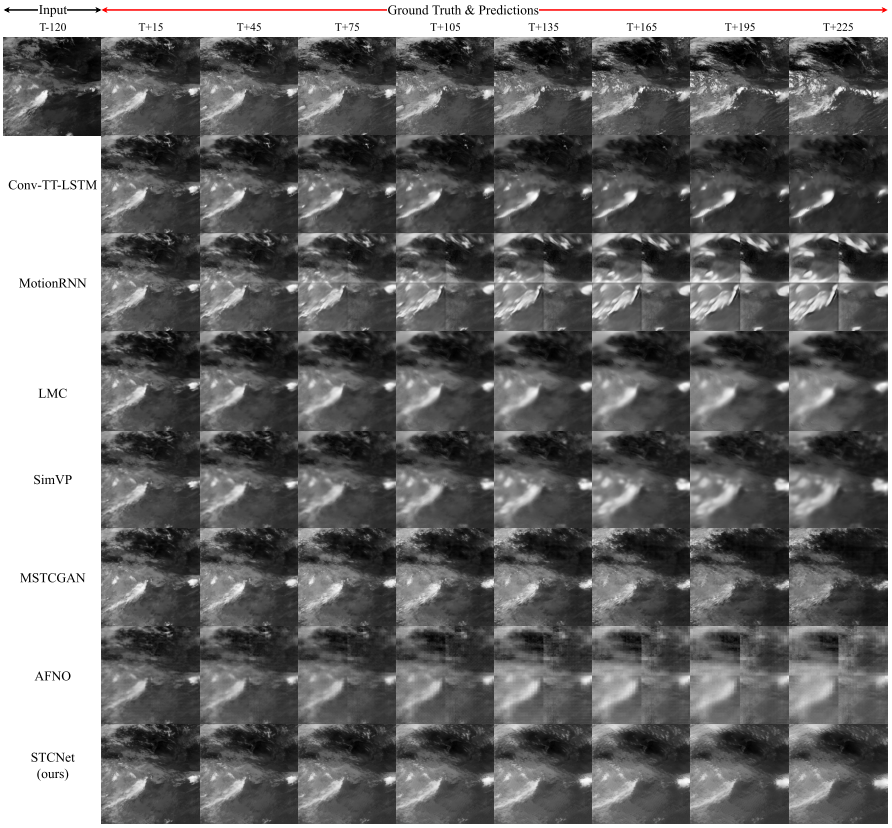

Fig. 8. 512 × 512-size predictions of baseline methods and STCNet. Every two frames are shown.

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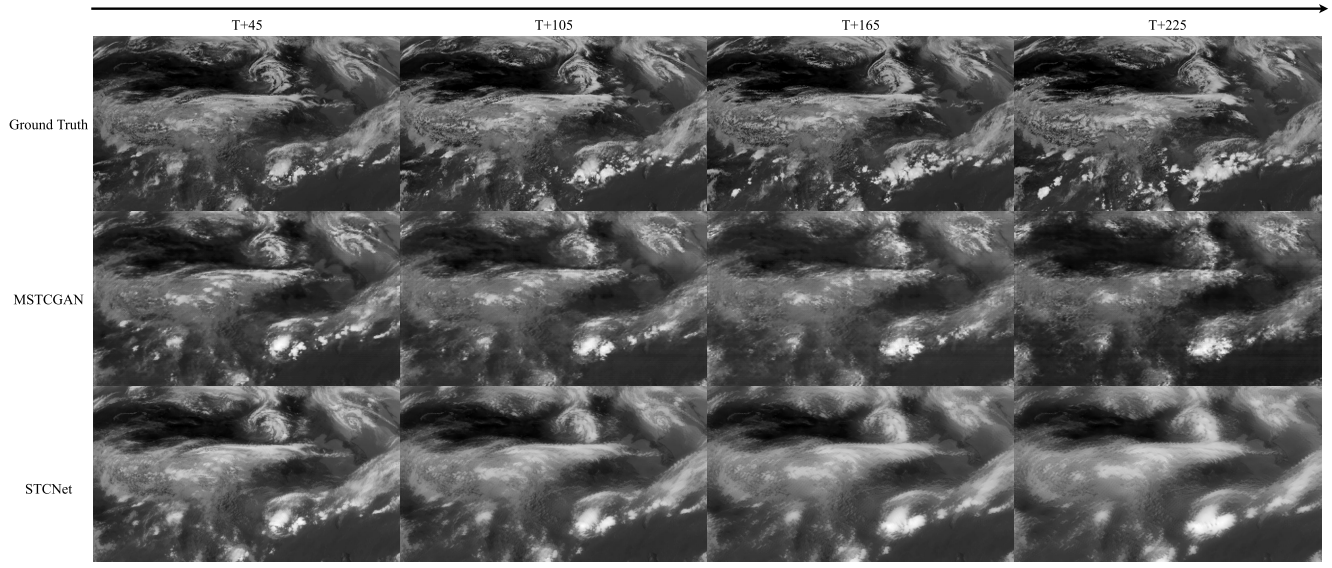

Fig. 9. 1280 × 730-size predictions of the baseline method MSTCGAN and STCNet. Every four frames are shown.

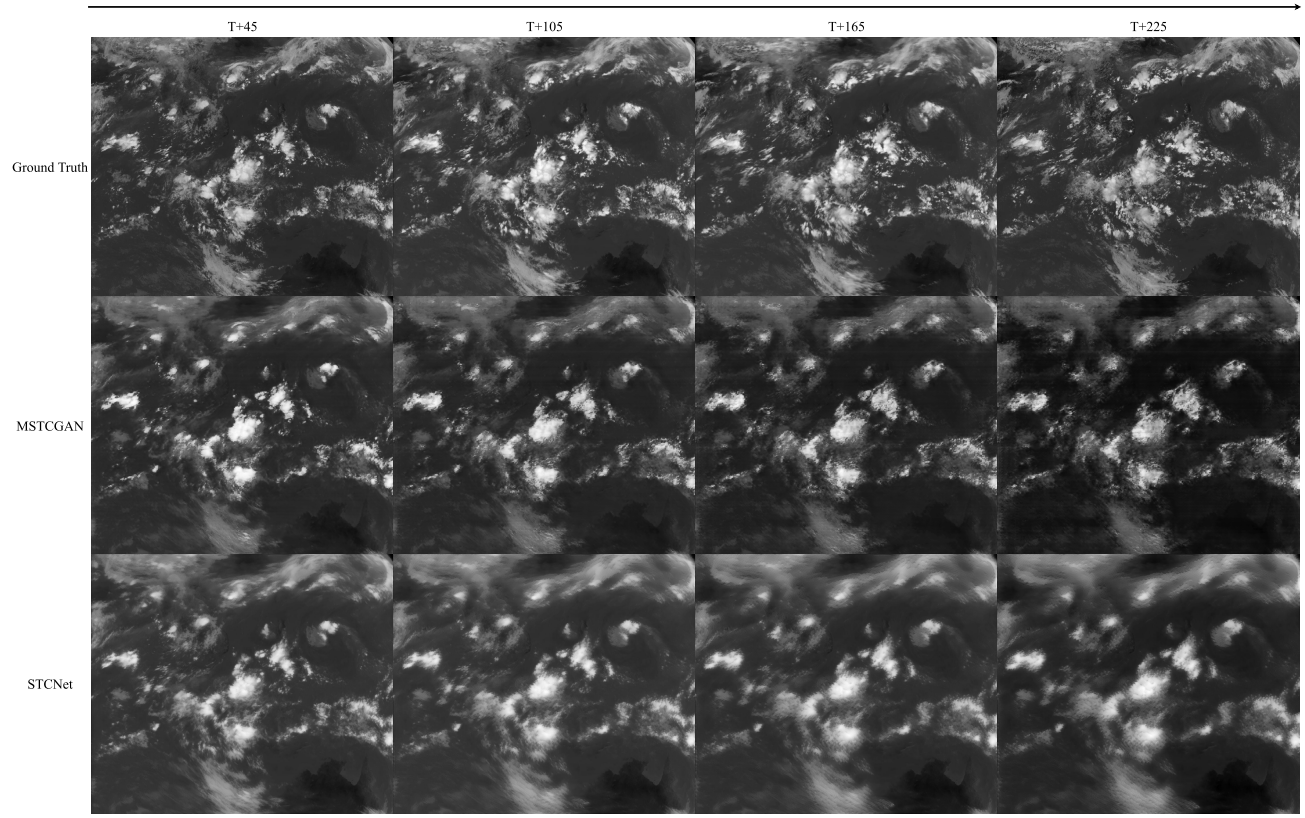

Fig. 10. 1750 × 1500-size predictions of the baseline method MSTCGAN and STCNet. Every four frames are shown.

Fifth, variant-5 significantly improves variant-4 in terms of PSNR, mse, and MAE scores, which shows that our designed uncertainty estimation mechanism enables the discriminator to characterize the temporal change of predicted sequences to make more accurate predictions. Finally, by comparing variant-5 and STCNet, we find that the scheduled sampling training strategy can further reduce the training difficulty and promote the performance of our proposed method. All the observations consistently demonstrate the effectiveness of key elements and our designed structures in the proposed STCNet.

D. Beyond Satellite Image Sequence Prediction

In this section, we conduct experiments beyond satellite image sequence prediction to evaluate its applicability and adaptability in real-world scenarios. Specifically, we first transfer our approach STCNet to datasets 2 and 3 to make

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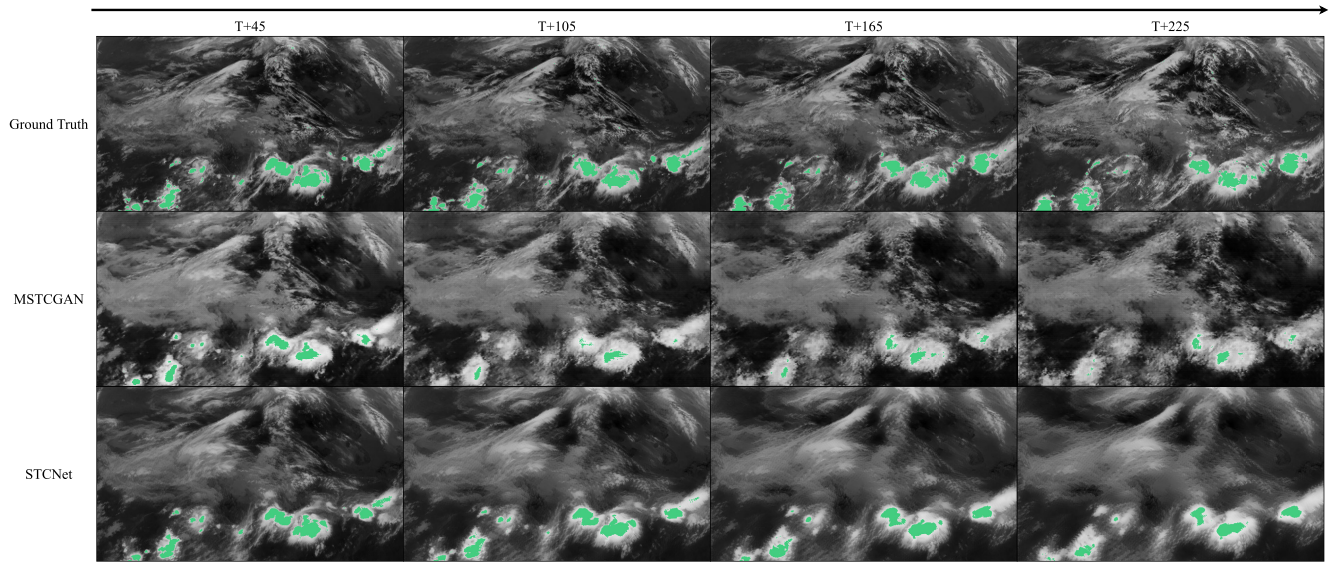

Fig. 11. Detected convection clouds (green parts) of predictions from the baseline method MSTCGAN and STCNet. Every four frames are shown.

TABLE VI
QUANTITATIVE TRANSFER PREDICTION RESULTS OF MSTCGAN AND OUR METHOD STCNET AND ON DATASET 2. THE BOLD DENOTES BETTER PERFORMANCE

Prediction Method	Performance $8 \rightarrow 16$				
	PSNR	MSE	MAE	GDL	LPIPS
MSTCGAN [25]	19.40	792.57	23.54	5.41	0.349
STCNet	**23.56**	**368.43**	**12.04**	**5.31**	**0.232**

TABLE VII
QUANTITATIVE TRANSFER PREDICTION RESULTS OF MSTCGAN AND OUR METHOD STCNET ON DATASET 3. THE BOLD DENOTES BETTER PERFORMANCE

Prediction Method	Performance $8 \rightarrow 16$				
	PSNR	MSE	MAE	GDL	LPIPS
MSTCGAN [25]	18.63	909.29	26.75	4.14	0.308
STCNet	**23.37**	**370.56**	**11.34**	**4.07**	**0.233**

predictions, so as to verify the spatial–temporal transfer ability of our method under more challenging scenarios. Then, to further validate the prediction accuracy of our method, we evaluate our proposed model on another significant task in the remote sensing domain, namely, convection cloud detection. Besides, considering the experiment cost, we select MSTCGAN as a baseline method, which is a competitive method both in terms of the quantitative results and visual effects.

1) Spatial–Temporal Transfer Prediction: Tables VI and VII show the transfer prediction results of our method STCNet and the baseline method MSTCGAN on datasets 2 and 3, respectively. Obviously, our proposed method significantly outperforms the baseline method MSTCGAN. Moreover, prediction samples of MSTCGAN and our method on the two datasets are shown in Figs. 9 and 10. Intuitively, we see that the baseline method MSTCGAN makes reasonable predictions in the first two hours but produces unnatural predicted frames in the last 2 h. In particular, motion of clouds in the last 2 h is odd and far from ground truth, which indicates that MSTCGAN cannot well maintain the motion coherency of the predicted sequence. Different from this, our model STCNet produces much more accurate predictions in terms of motion patterns. The quantitative and qualitative results consistently demonstrate the advantage of our method for spatial–temporal transfer prediction.

2) Convection Cloud Detection: We conduct the cloud detection task on predictions from our proposed method STCNet and the baseline method MSTCGAN by using a pretrained UNet [58] model, which is trained on a convection cloud detection dataset manually labeled by us. As shown in Fig. 11, the convection clouds (green parts) from predictions produced by MSTCGAN fail to match with the ground truth in terms of shape, size, and location. Besides, we find that the motion patterns of the convection clouds are not coherent. In particular, there exists an abrupt dissipation of convection clouds between the previous frame ($T + 45$) and the next frame ($T + 105$). Compared with MSTCGAN, our approach more accurately predicts the convection clouds, and the motion dynamics of the predicted convection clouds are more coherent. The experiments on the convection cloud detection task again verify the superiority of our method STCNet.

V. CONCLUSION

In this article, we propose a novel and effective method STCNet for satellite image sequence prediction. In our approach, a multilevel motion memory-based predictor and a time-variant frame discriminator are proposed to maintain temporal consistency and enhance spatial consistency, respectively. In addition, a scheduled sampling strategy is proposed to enhance the training stability of our proposed method. Comprehensive experiments are conducted on satellite image sequences from the FY-4A meteorological satellite, which demonstrates the effectiveness and superiority of our proposed methods. We hope that our work can inspire more researchers to study the satellite image sequence prediction problem.

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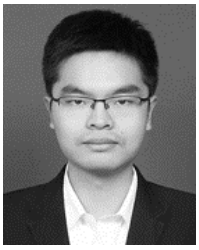

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